

Neural network Method: Loss Minimization control of a PMSM with core Resistance Assessment

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Abstract— Permanent magnet synchronous motors (PMSMs) are often used in industry for high-performance applications. Their key features are high power density, linear torque control capability, high efficiency, and fast dynamic response. Today, PMSMs are prevalent especially for their use in hybrid electric vehicles. Since operating the motor at high efficiency values is critically important for electric vehicles, as for all other applications, minimum loss control appears to be an inevitable requirement in PMSMs. In this study, a neural network-based intelligent minimum loss control technique is applied to a PMSM. It is shown by means of the results obtained that the total machine losses can be controlled in a way that keeps them at a minimum level. It is worth noting here that this improvement is achieved compared to the case with I_d set to zero, where no minimum loss control technique is used. Within this context, hysteresis and eddy current losses are primarily obtained under certain conditions by means of a PMSM finite element model, initially developed by CEDRAT as an educational demo. A comprehensive loss model with a dynamic core resistor estimator is developed using this information. A neural network controller is then applied to this model and comparisons are made with analytical methods such as field weakening and maximum torque per ampere control techniques. Finally, the obtained results are discussed.

Keywords — Permanent magnet synchronous motor, energy efficiency, neural network, loss model

I. INTRODUCTION

When working with limited energy sources, as in electric cars, it is obvious that loss of power has a great effect on overall efficiency and performance. Therefore, it is of the utmost necessity that the losses be reduced as much as possible to prevent the limited power sources of the automobiles from being drained. When reviewing the literature, it is observed that there are many studies on the investigation of losses in permanent magnet synchronous motors (PMSMs). The authors of [1, 2] considered only the copper losses. Neglecting core losses is the main shortcoming of this work.

To achieve a complete investigation of losses on PMSMs, several authors conducted their research considering iron losses along with copper losses. In electric vehicle applications, motors are generally required to operate in wide speed ranges. The authors of [5,6,7,8] performed analyses on iron losses and/or equivalent core loss resistance under fixed rotor speed operation conditions. In fixed speed operations, frequency remains fixed, too. This introduces some simplicity and easiness, especially in loss calculations with frequency-dependent partial differential equation components. However, this type of analysis has limited use and cannot be used in variable speed operations. Furthermore, there is some research in which speed variation

was taken into account. For example, in [9-16], the authors discussed different particular speed values, but they conducted their research with a fixed core resistance. In these cases, they neglected the variation of core resistance, which is related to frequency. This approach provides advantages in calculations, yet causes certain errors due to neglecting frequency effects. The authors of [17, 18,19,20] also examined core resistance changes in variable speeds. These authors, like those mentioned above, ran their analyses only for certain speed values; consequently, they neglected speed variations between the selected speed values. Finally, the authors of [21,22] investigated dynamic changes in core resistance with variable speeds by using highly complex analytical calculations. Besides determining the iron core resistance values, another important issue in the literature concerns the definition of the loss minimization technique.

In electric car drive systems, the amplitude of the voltage of the vehicle battery terminals tends to change under different loads. In the low voltage range, despite the lack of sufficient rotor speed, a force requirement arises for the control mode to enter the field weakening area in order to achieve the desired operating speed. Thus, field weakening (FW) control is very important for electric vehicles. The FW technique was used in many studies to extend the operating speed

range. FW applications were examined and a new analytical method was proposed by author. The results of the proposed method were compared to those of the earlier methods. In that paper, highly complex analytical calculations were used and only the FW region was examined. A maximum torque per ampere (MTPA) strategy was used to reduce motor losses. These studies were also based on a single method that included intensive analytical procedures, as in the previous research. In addition to studies that focused only on one method, there exist several studies [1,5,12] that use the FW and MTPA strategies jointly. In these studies, the increase in performance using a combination of the two methods is apparent. The methods presented in [1,12,] yield good results but require intensive and complex analytical procedures. To reduce this complexity, the authors of [5] used look-up tables in order to keep procedures as simple as possible. However, the construction of tables is still very time-consuming for researchers.

In this paper, a PMSM loss model was built, which can be used in designing a controller to minimize losses and thus increase the efficiency in PMSM. The model does not neglect electrical losses. A neuro-fuzzy-based core resistance estimator for an interior permanent magnet synchronous motor has been designed. This estimator was used for estimating the core resistance dynamically as the speed changes. Consequently, a neural network controller was designed and applied to the motor to decrease its losses. At the end, the results were analyzed.

II. PMSM LOSS MODEL

Primarily, it is necessary to set up a PMSM loss model that does not neglect losses in order to create a minimum loss control simulation in the PMSM. The PMSM dq-reference model was selected as the baseline in this paper, since it is the most appropriate for the control method applied. Saturation effects and inverter switching losses are neglected in this work. Equivalent circuits of the d and q axes of the PMSM, which do not neglect losses, are presented in Figures 1 and 2 [16].

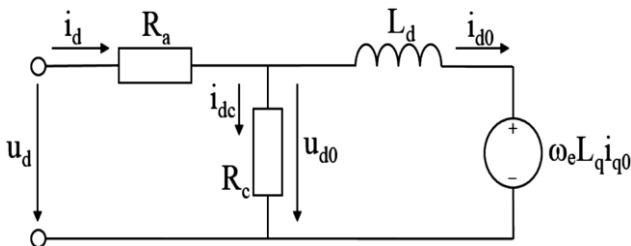


Fig.1. The d-axes equivalent circuit of a PMSM with core resistance

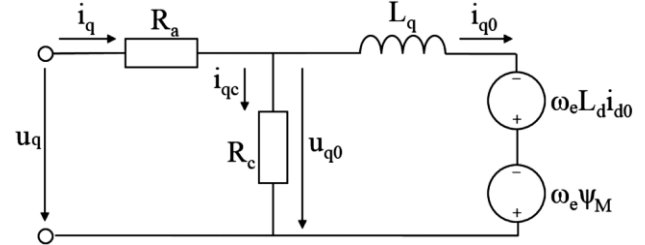


Fig.2. The q-axes equivalent circuit of a PMSM with core resistance

Of the resistors shown in the circuit, R_c stands for equivalent iron core resistance and R_a represents phase winding resistance. Currents i_d and i_q represent the stator d and q axes' current components, i_{dc} and i_{qc} the stator d and q axes' iron core current components, and i_{d0} and i_{q0} the remaining d and q axes' current components. In addition, u_d and u_q represent the d and q axes' voltages, L_d and L_q represent the inductances along the d and q axes, and M represents the ux density of the permanent magnets. Instantaneous equations, which represent the PMSM where the losses are also taken into account, can be derived from equivalent circuits shown in Figures 1 and 2, as follows:

$$\frac{di_{d0}}{dt} = \frac{1}{L_d} \left(R_c \frac{u_d + R_c i_{d0}}{R_a + R_c} - R_c i_{d0} + \omega_e L_q i_{q0} \right)$$

$$\frac{di_{q0}}{dt} = \frac{1}{L_q} \left(R_c \frac{u_q + R_c i_{q0}}{R_a + R_c} - R_c i_{q0} + \omega_e (\Psi_M + L_d i_{d0}) \right)$$

The impact of the iron core resistance, R_c , on currents can be clearly observed in Eqs. (1) and (2). The generated torque expression of the PMSM adapted to a dq equivalent circuit can be obtained as presented in Eq. (3), where p represents the number of pole pairs:

$$T_e = \frac{3}{2} p [\Psi_{PM} i_{q0} + (L_d - L_q) i_{d0} i_{q0}]$$

Finite element analysis of core losses in PMSM

In loss minimization studies of PMSMs, identifying the iron losses of the motor has great significance in a loss minimization procedure. Analytical methods trying to solve this problem appear quite difficult, because loss equations include partial differential equations. After investigating some previous studies [4,7,8,13,17,20], the usage of a motor model developed by the finite elements method (FEM) was chosen as the most suitable solution to the problem.

The FEM is a method used for the solutions of Laplace- and Poisson-type partial differential equations.

Using the FEM, the uxes, generated torque, and iron losses of the PMSM can be calculated in detail once the physical

size of the motor and the parameters of the materials used are introduced into the software.

In this study, FEM analysis is carried out with a premade educational demonstration FEM, which was created and released by CEDRAT and embedded in Flux 2D software. For more information about the FEM model, please visit <http://www.cedrat.com/>; the analysis was carried out under a registered trial license.

Consequently, the hysteresis power loss (W_h) and the eddy current power loss (W_e) defined by Eqs. (4) and (5), respectively [6], as the two main components of total iron losses (W_{fe}), are found by running FEM analysis at different speeds. The results obtained are shown in Fig.3.

$$W_h = k_h [1 + c(r - 1)] B^2 f$$

Here B is the peak value of the magnetic flux density, k_h is the material constant, c is the magnetic flux density ratio B_{min}/B_{max} , r is the empirical factor, and f is the fundamental frequency.

$$W_e = k_{ec} \sum_{n=1}^{\infty} B_n^2 (nf)^2$$

B_n is the peak value of the magnetic flux density for the n th harmonic order. Here, if the skin effect of the eddy current is neglected, k_{ec} can be written as follows:

$$k_{ec} = \frac{\pi^2 d^2}{6 \rho_e \rho}$$

Here d is the sheet thickness, ρ is the sheet density, and ρ_e is the specific electrical resistance of the steel. From Fig.3, it can be clearly seen that the variation of the eddy current losses has a quadratic shape that responds to a change in frequency, as defined in Eq. (5). Additionally, the change of the hysteresis loss has a linear shape that suits the definition given in Eq. (4). It is possible to calculate the iron core equivalent resistance, R_c , by means of the total iron losses obtained in Fig.3. In this study, the preformed lossless model was operated at each value of speed given in Fig.3, using the $I_d = 0$ control method to obtain corresponding u_o voltage values. Then the u_o voltage values obtained from the lossless model were used in Eq. (8) together with the W_{fe} values obtained from FEM analyses in order to find the iron core equivalent resistances, R_c , for each speed point, as shown in Fig.4. Eq. (8) can be derived from the d and q axes' equivalent circuits, given in Figures 1 and 2 [14]. The other parameters of the PMSM are given in the Table.

Table. Other PMSM parameters.

Parameter [unit]	Value
p_p [-]	4
R_a [Ω]	0.42
Ψ_{PM} [Wb]	0.5052
J [kgm^2]	0.0637
L_d [mH]	2.11e-3
L_q [mH]	8.98e-3

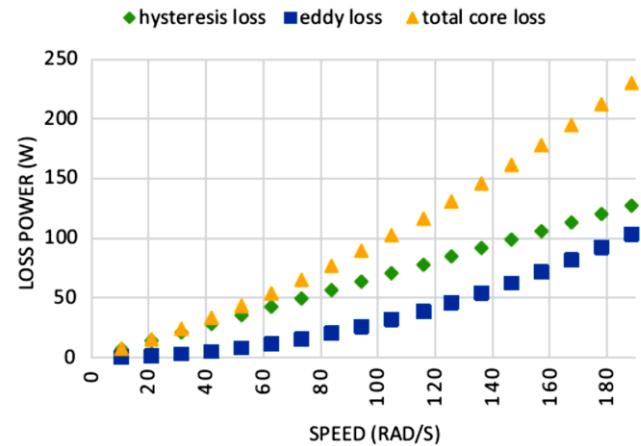


Fig.3. Core losses on the PMSM

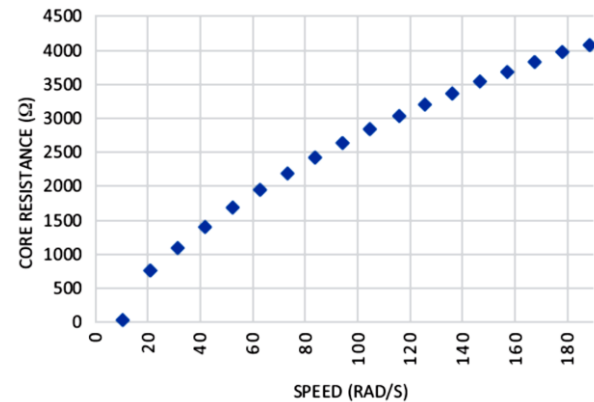


Fig.4. Core resistance derivation vs. speed

$$u_0^2 = u_{d0}^2 + u_{q0}^2$$

$$W_{fe} = W_h + W_e = \frac{\omega_e^2 (\rho L_q i_{q0})^2}{R_c} + \frac{\omega_e^2 (L_d i_{d0} + \psi_M)^2}{R_c} = \frac{u_0^2}{R_c}$$

Core resistance estimation with adaptive neuro-fuzzy interference system

For carrying out loss analysis on variable speed drive systems, core resistance values for all corresponding speed values must be known. The FEM cannot be used in dynamic analysis because FEM models are required to work with constant values. In this regard, intelligent systems are

emerging as the most appropriate solution to dynamically determine the core resistance. The results obtained by FEM using fixed-speed simulations can be used as the required training data for developing a neuro-fuzzy intelligent prediction model. In the case of limited access to the required training data, results obtained by ANN may not be satisfactory. In such a case, a fuzzy logic controller can be used as an auxiliary element to improve the results obtained by an ANN controller.

III. PMSM MINIMUM LOSS CONTROL WITH NEURAL NETWORK

Technological developments and advances in artificial intelligence applications have allowed modern analytical methods to replace analytical solutions. As a result of the studies, it is observed that modern methods are just as successful as classical methods in the analysis of electrical machinery; thus, iterative applications that require a great deal of calculations are replaced by trained systems.

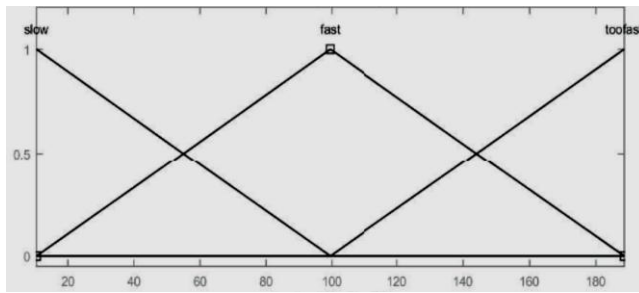


Fig.5. Membership functions for core resistor estimator

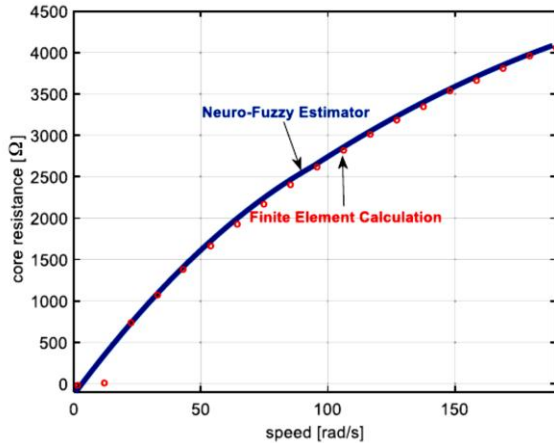


Fig.6. Comparison of estimated core resistance values obtained from finite element results to those obtained from the neuro-fuzzy estimator

For control actions based on analytical calculations, it is necessary to compare and contrast many parameters again and again and generate reference currents using many parameters again. However, by utilizing the designed neural network, it is possible to generate desired reference currents

using a small number of parameters and without complex calculations after the training procedure. In this context, when designing a neural network controller for the present problem, it was primarily developed as an analytical loss minimization controller model. This controller was initially developed in a previous study in [12] as a combination of the MTPA and FW control methods. The operation of this controller can be summarized by the flow diagram given in Fig.7.

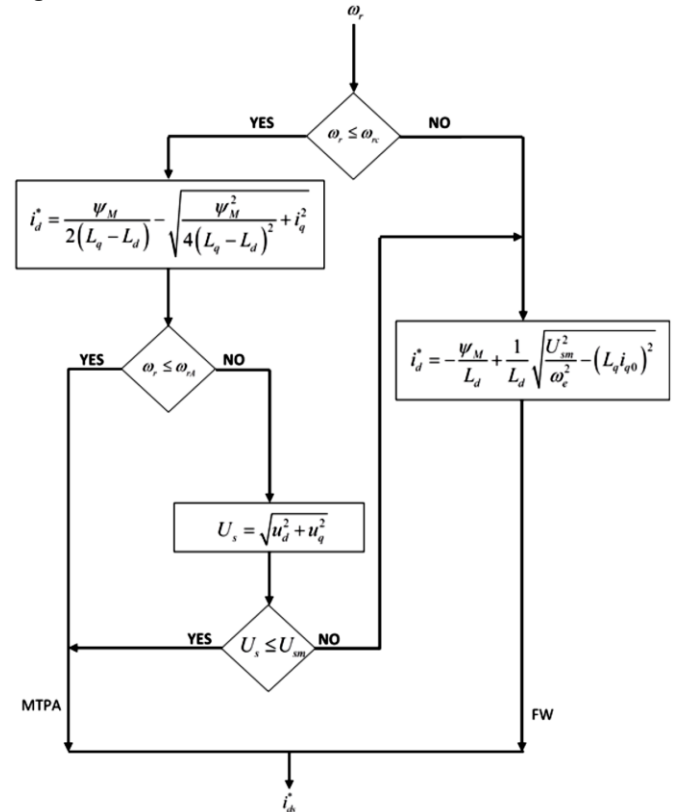


Fig.7. MTPA and FW controller flow chart.

As shown in Fig.8, the design was developed comprising two layers and was planned to have 10 neurons in the hidden layer.

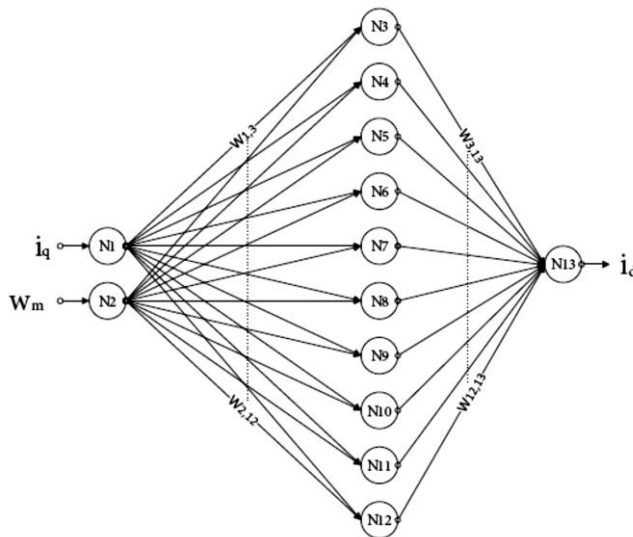


Fig.8. Design of neural network minimum loss controller

Training was conducted with the Levenberg {Marquardt} backpropagation method. The outputs obtained in the training are shown in Fig.9.

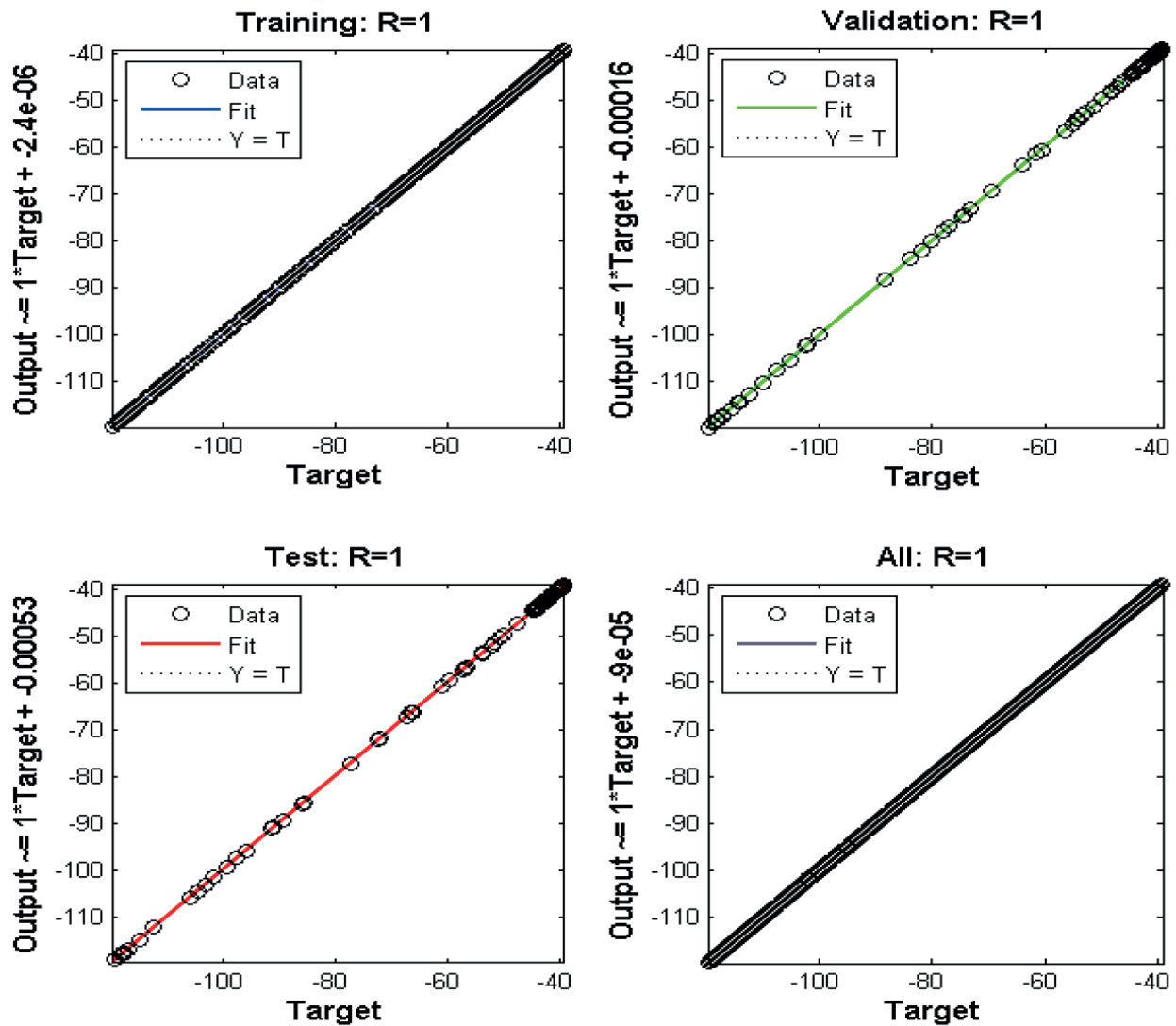


Fig.9. Neural network training results.

The simulation model, including the designed neural network controller, was developed as described in Fig.10. Additionally, an operation scenario was applied to the simulation. In this scenario, during the first 0.2 s, the motor is operated at 70 rad/s constant speed value, which is below the nominal speed (94.28 rad/s) of the motor. Then, in order to examine the effects of speed changes on motor behavior between 0.2 and 0.3 s, the speed is increased linearly with a 550 rad/s² slope value. Between 0.3 and 0.45 s, the slope

value is decreased to 125 rad/s². In addition to examining the motor's dynamic response to load changes, a dynamic load torque, whose value is defined as $TL = 3um$, is applied to the motor from the beginning of the operation. To make an overall assessment, the results obtained are given in Figures 11{18. The results are compared to those of the MTPA and FW control methods.

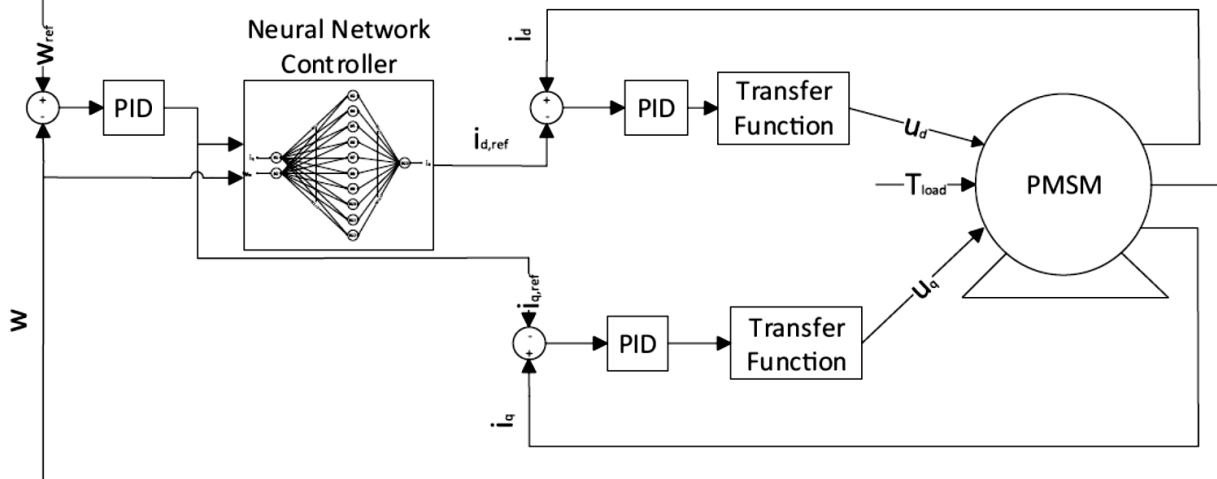


Fig.10. PMSM minimum loss control simulation model.

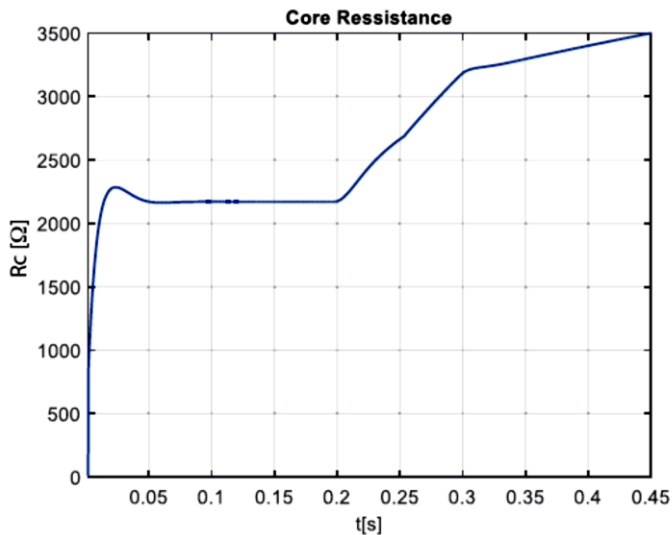


Fig.11. Core resistance estimation during an operation scenario with neural network and MTPA and FW methods.

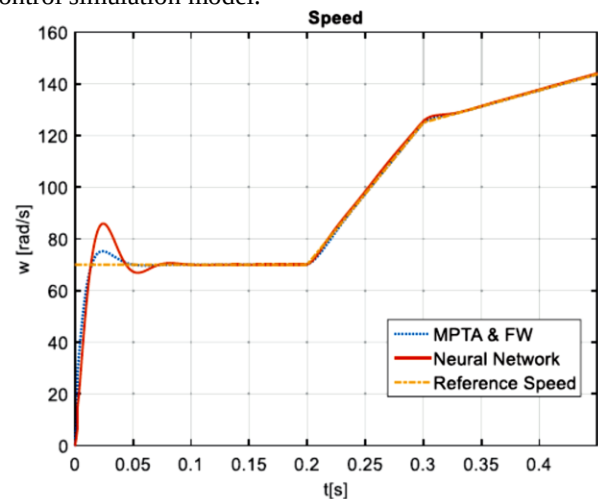


Fig.12. Comparison of rotor speed responses between neural network and MTPA and FW control methods.

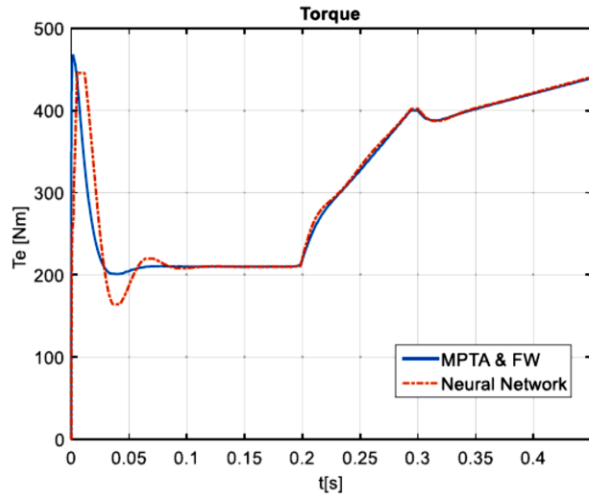


Fig.13. Comparison of generated torque responses between neural network and MTPA and FW control methods.

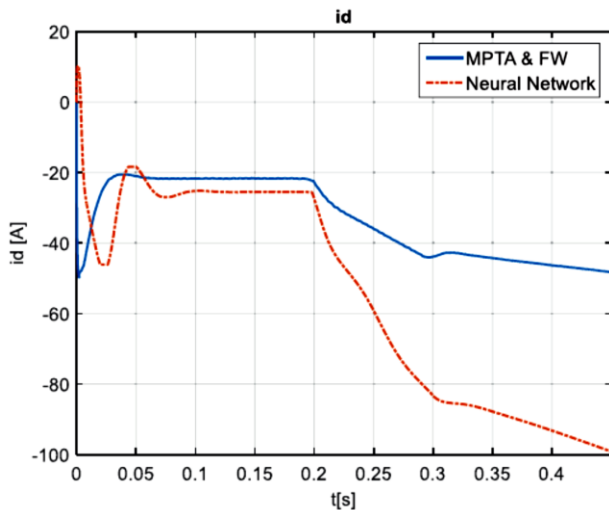


Fig.14. Comparison of d axes component current variation, i_d , between neural network and MTPA and FW control methods.

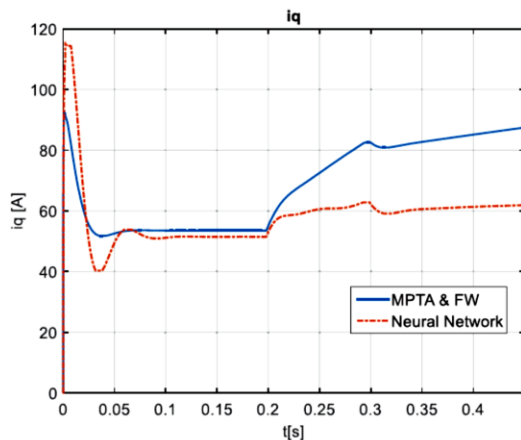


Fig.15. Comparison of q axes component current responses, i_q , between neural network and MTPA and FW control methods.

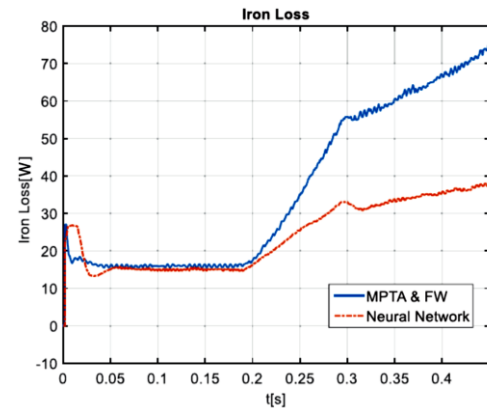


Fig.16. Comparison of iron power loss variation between neural network and MTPA and FW control methods.

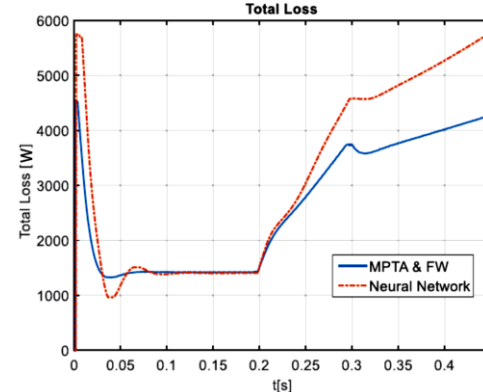


Fig.17. Comparison of total power loss variations between neural network and MTPA and FW control methods.

The dynamic change of the core resistance is given in Fig.11. When the torque and rotor speed variations given in Figures 12 and 13 are examined, it is observed that the motor variables follow their references with great success and show a good dynamic response. Besides, the neural network controller successfully adjusts the values of the d and q currents, as can be seen from Figures 14 and 15. Consequently, the applied neural network controller clearly produces lower core losses than the MTPA and FW controller, as can be seen from Fig.16. When Fig.17 is investigated in detail around the high-speed region, it can be observed that the MTPA and FW controller causes lower values of total power loss, which is the sum of the iron losses and copper losses; however, in the low-speed region, the neural network controller becomes more advantageous. This situation can be clearly seen in efficiency derivation as given in Fig.18. Again, in the low-speed region, the neural network is more advantageous than the MTPA and FW.

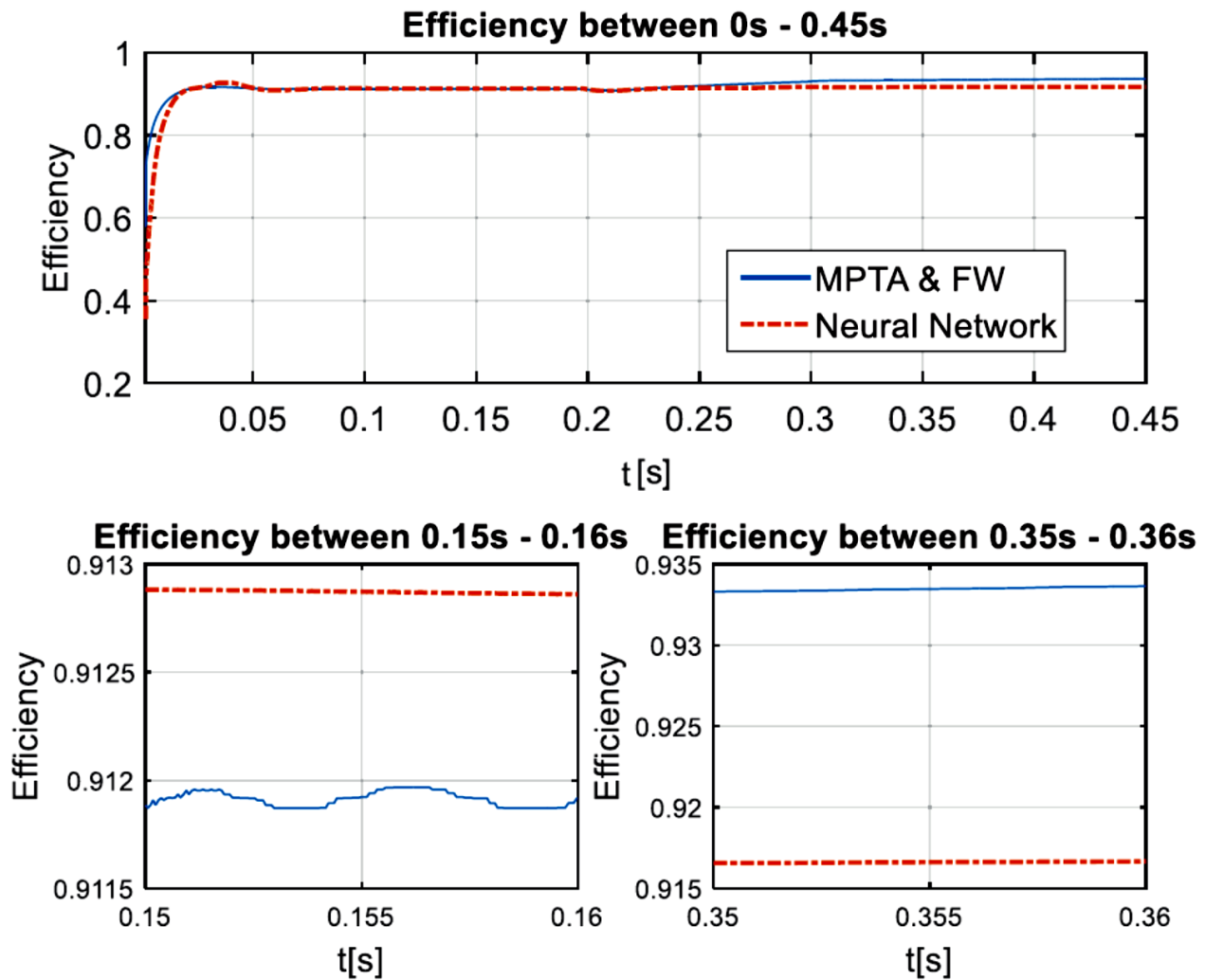


Fig.18. Comparison of motor efficiency variations between neural network and MTPA and FW control methods.

IV. CONCLUSION

This paper carried out a study on minimizing the electrical losses of the PMSM with intelligent systems. In this context, as a novel approach, a neuro-fuzzy-based estimator was designed in order to eliminate an important deficiency found in the literature about determining the dynamic core resistance. Once the dynamic core resistance is available, it is possible to reduce the core losses of the machine by means of a controller. For this reason, a neural network controller, which is a novel approach employing the dynamic core resistance, was designed and applied to the motor. The simulation results show clearly that minimum loss control can be obtained by the proposed control technique.. Finally, the obtained results were compared to the results of the MTPA and FW control methods. Because iron loss in the

constant torque zone increases in direct proportion to speed increase, the effects of reducing iron loss assure better results in the high-speed zone; this can be seen clearly in Fig.16.

V. REFERENCES

- [1] B. Asaei, B. Rahrovi, "Minimum-copper-loss control over full speed range of an IPMSM drive for hybrid electric vehicle application", IEEE 2010 Vehicle Power and Propulsion Conference, NY, pp. 1-6, 2010.
- [2] YS. Jeong, SK. Sul, S. Hiti, KM Rahman, "Online minimum-copper-loss control of an interior permanent-magnetsynchronous machine for automotive applications", IEEE Transactions on Industry Applications, Vol.42, Issue.2, pp.1222-1229, 2006.
- [3] LT. Huynh, Hung Thang Phung, Toai QuangTon, "Using Convolutional Neural Network to Recognize Handwritten Digits",

- International Journal of Computer Sciences and Engineering, Vol.3, Issue.5, pp.203-206, 2015.
- [4] M.H. Babu, D.S.S.N. Raju, A. Sarvani, M.P. Rao, "Scattering of Sodar Signal by Turbulence in Homogenous Isotopic and other Mediums", International Journal of Computer Sciences and Engineering, Vol.2, Issue.4, pp.1-5, 2014.
- [5] H. Aorith, J. Wang, P. Lazari, "A new loss minimization algorithm for interior permanent magnet synchronous machine drives", IEEE 2013 International Electric Machines and Drives Conference, NY, pp.526-533, 2013.
- [6] MH. Gracia, E. Lange, K. Hameyer, "Numerical calculation of iron losses in electrical machines with a modified post-processing formula", In: Proceedings of the 16th International Symposium on Electromagnetic Fields COMPUMAG, Germany, pp. 1-2, 2007.
- [7] SH. Han, WL. Soong, TM. Jahns, "An analytical design approach for reducing stator iron losses in interior PM synchronous machines during ux-weakening operation", IEEE 2007 Industry Applications Annual Meeting, LA, pp. 103-110, 2007.
- [8] G. Ugalde, G. Almandoz, J. Poza, A. Gonzalez, "Computation of iron losses in permanent magnet machines by multi-domain simulations", European Conference on Power Electronics and Applications, Spain, pp.1-10, 2009.
- [9] FF. Bernal FF, A. Garefa-Cerrada, R. Faure, "Loss-minimization control of synchronous machines with constant excitation", IEEE 1998 Annual Power Electronics Specialists Conference, Japan, pp.132-138, 1998.
- [10] C. Cavallaro, AO. Tommaso, R. Miceli, A. Raciti, GR. Galluzzo, M. Trapanese, "Analysis a DSP implementation and experimental validation of a loss minimization algorithm applied to permanent magnet synchronous motor drives", 29th Annual Conference of the IEEE Industrial Electronics Society, Roanoke, pp. 312-317, 2003.
- [11] C. Cavallaro, AO. Tommaso, R. Miceli, A. Raciti, RG. Galluzzo, M. Trapanese, "Efficiency enhancement of permanent-magnet synchronous motor drives by online loss minimization approaches", IEEE Transactions on Industrial Electronics, Vol.52, Issue.4, pp.1153-1160, 2005.
- [12] S. Morimoto, Y. Tong, Y. Takeda, T. Hirasa, "Loss minimization control of permanent magnet synchronous motor drives", IEEE Transactions on industrial electronics, Vol.41, Issue.5, pp.511-517, 1994.
- [13] S. Agrawal, K.D. Kulat, M. B.Daigavane, "Evaluation of Routing Algorithm for Ad-hoc and Wireless Sensor Network Protocol", International Journal of Computer Sciences and Engineering, Vol.1, Issue.2, pp.11-18, 2013.
- [14] S. Morimoto, Y. Tong, Y. Takeda, T. Hirasa, "Loss minimization control of permanent magnet synchronous motor drives", IEEE Transactions on industrial electronics, Vol.41, Issue.5, pp.511-517, 1994.
- [15] MN Uddin, S. Member, H. Zou, F. Azevedo, "Online loss minimization based adaptive ux observer for direct torque and ux control of PMSM drive", IEEE 2014 Industry Applications Society Annual Meeting, Canada, pp.1-7, 2014.
- [16] Y. Watanabe, TW. Kim, A. Mushi, A. Kawamura, "Research on overall efficiency improvement of electric vehicles by MTHDPAM control method", IEEE 2012 15th International Power Electronics and Motion Control Conference, Serbia, pp.5-12, 2012.
- [17] A. Krings, S. Nategh, O. Wallmark, J. Soulard, "Local iron loss identification by thermal measurements on an outer-rotor permanent magnet synchronous machine", IEEE 2012 15th International Conference on Electrical Machines and Systems, Japan, pp.1-5, 2012.
- [18] R. Dutta, L. Chong, FM. Rahman, "Analysis and experimental verification of losses in a concentrated wound interior permanent magnet machine", Progress in electromagnetics Research B, Vol.48, Issue.1, pp. 221-248, 2013.
- [19] I. Shchur, A. Rusek, O. Makarchuk, M. Mandzyuk, "Definition of parameters of mathematical model of pmsm for electric vehicles on the basis of computer and experimental research", Maszyny Elektryczne Zeszyty Problemowe, Vol.1, Issue.101, pp.147-152, 2014.
- [20] YK. Chin, J. Soulard, "Modeling of iron losses in permanent magnet synchronous motors with field-weakening capability for electric vehicles", International Journal of Automotive Technology, Vol.4, Issue.2, pp.87-94, 2003.
- [21] JY. Lee, SH. Lee, GH. Lee, JP. Hong, J. Hur, "Determination of parameters considering magnetic nonlinearity in an interior permanent magnet synchronous motor", IEEE Transaction on Magnetics, Vol.42, Issue.4, pp.1303-1306, 2006.
- [22] BH. Lee, SO. Kwon, T. Sun, JP. Hong, GH. Lee, J. Hur, "Modeling of core loss resistance for d-q equivalent circuit analysis of IPMSM considering harmonic linkage ux", IEEE Transaction on Magnetics, Vol.47, Issue.5, pp.1066-1069, 2011.